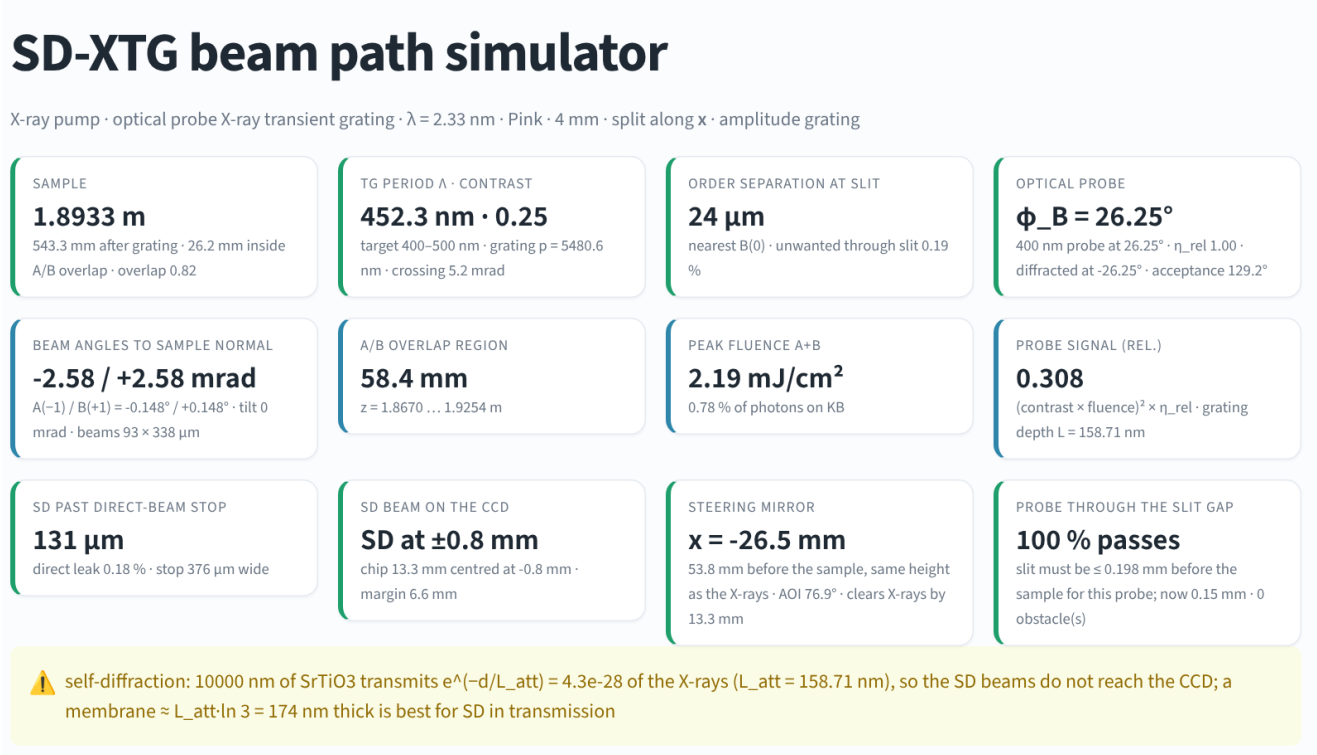
SD-XTG beam-path simulator

User guide for testers · version 2026-10-22 r12 · PAL beamtime, O K-edge (λ = 2.33 nm) · D. Fainozzi

What it is for

We want a transient grating written by the FEL itself, and we would rather not discover at 3 am, during the beamtime, that the slit eats the probe. The KB focuses the beam (HM f = 2 m, VM 0.5 m downstream with f = 1.5 m, ~40 µm focus on the CCD). A bar beamstop after the KB cuts it into two converging half-beams, **A** and **B**. A grating diffracts both, and the 1st orders **A(−1)** and **B(+1)** cross on the sample and write the TG. A slit just before the sample keeps only these two orders. Then either the self-diffracted X-rays ($2k_A - k_B$, $2k_B - k_A$) go to the CCD past a direct-beam stop (**SD mode**), or an optical beam reads the grating (**X-ray pump · optical probe mode**).

The simulator traces all of it for the pink (4 mm) and mono (1 mm) beams. It tells us where the sample goes, which grating period gives the TG period we want, and whether slit, beamstops, CCD, steering mirror and probe still agree with each other. Every number and every drawing is recomputed when anything changes. All views come from the same rays, so if two views disagree it is a bug and we want to hear about it.



Top of the window, pink beam, optical-probe mode. Green cards: fine. Red cards: something fails, and the card says what. Yellow boxes: physics we should not ignore (here: 10 µm of STO absorbs the SD beams completely).

What is new in this version

Feature	What it does
TG period first	Default target Λ = 400–500 nm, setpoint 450 nm. The grating period is solved from it (pink 5481 nm at z_g = 1.35 m, mono 1138 nm at z_g = 1.50 m).
Beam size = waist w_0	4 mm / 1 mm are 1/e² radii (FWHM = 1.177 w_0). Switchable to 1/e² diameter or FWHM in the sidebar.
Optical probe in the X-ray plane	Always horizontal, same height as the X-rays, so it diffracts in the same plane as SD. SD beams stay on in probe mode.
Phase-matched probe	Angle from $\sin \varphi_B = \lambda_{probe}/(2\Lambda)$, formula with numbers in the GUI; every drawing uses it.
Probe waist	w_0 adjustable, default 50 µm. Spreads after the sample with the Rayleigh range.
Probe through the slit	Fraction of the probe through the gap, and the largest slit → sample distance. Default slit now 0.15 mm before the sample.

Feature	What it does
Probe on the CCD plane	To-scale map of the CCD plane: chip, SD spots, where the probes land. Warns if a probe hits the X-ray chip.
Sample material	Formula, density, thickness, n. Henke L_{att} from `periodictable`. Inputs flagged **consistent** / **not consistent**.
TG in the sample	Absorbed-energy map in x–z, depth profile, eV/atom and a Dulong–Petit ΔT.
SD through the sample	Exact SD angle, phase mismatch and $e^{-d/L}$ transmission. Best SD thickness $L_{att}\cdot\ln 3$.
3D view	Whole beamline, grating → CCD to scale, pump–probe close-up to scale. Alt/⌥ + scroll zooms to the cursor; the camera is remembered.
One outline everywhere	Every beam is drawn at its 1/e² radius, A and B are cut from the full beam, `check_views.py` verifies it.

Installing and starting it

1. Unzip `sdxtg_simulator.zip` anywhere (or clone github.com/D3nFaina/xtg_and_sdxtg-simulator).
2. Install what it needs, once, in the Python that Spyder uses (Anaconda Prompt, or `!pip …` in the Spyder console):

```
pip install streamlit plotly numpy matplotlib periodictable
```

3. In Spyder open `run_app.py` (or `sdxtg_app.py`) and press **Run (F5)**. From a terminal: `python run_app.py`. The browser opens on the GUI. Streamlit ≥ 1.30 works; we tested 1.38 and 1.66.
4. Check the sidebar: it says **Version 2026-10-22 r12** and the folder the code runs from. Every launch starts a new server on the next free port (8501, 8502, …), so an old browser tab happily keeps showing the old version. Close old tabs. If the version line is missing, you are testing the past.
5. The server log goes to `sdxtg_app.log` next to the code. Send it with any bug report.

How we use it

1. **Pick the beam** (pink / mono) and the **experiment** (X-ray SD on CCD / X-ray pump · optical probe). Switching back and forth keeps every value.
2. **Set the TG target**: Λ_{min}–Λ_{max} and the setpoint. Settings outside the range are discarded (red card, grey scan pixels).
3. **Set the sample**: preset or formula, density, thickness, n. Read the flags before trusting anything else.
4. **Set the probe** (probe mode): wavelength, waist, steering-mirror distance and size. Leave Probe at the Bragg angle on unless you enjoy losing signal.
5. **Move the elements**: beamstop, grating, slit, direct-beam stop, CCD. The cards say what breaks.
6. **Scan**: Correlation scan → ▶ Run scan. Click a pixel or apply the best point.
7. **Check with waves** in the wave-optics tab, then **save** parameters (JSON), beam sizes and scans (CSV).

Tab	What we see
Beamline	HM → CCD sketch and a grating → CCD zoom: every order, slit, sample, A/B overlap region, stops, probe arrows.
3D view	Whole beamline (transverse axes stretched), grating → CCD to scale with mirror and probe, pump–probe close-up to scale.
Beam size	FWHM vs z on both axes for the full beam, A and B; sizes at every element (CSV).
Slit · sample · CCD	Profiles at the slit, sample and direct-beam stop with the probe on top; SD spots on the CCD; the to-scale CCD plane with the probes.
Angles · optical probe	Angles to the sample normal, the phase-matching formula with numbers, η_{rel} vs angle, pump–probe close-up, top view of the probe path.
TG in the sample	Absorbed energy in x–z, depth profile, eV/atom, ΔT.
Correlation scan	Grating z × Λ maps (12 quantities), feasible region outlined, definition printed under each map.
Wave-optics check	Fresnel propagation: wave vs ray profiles, TG fringes, contrast.

Tab	What we see
Model & assumptions	Geometry, formulas, and the placeholders we still have to replace.

The TG period runs the show

$\Lambda = \lambda / (2 \sin(\theta/2))$, with θ the angle between A(−1) and B(+1). The grating period sets θ almost alone ($\theta \approx 2\lambda/p + \theta_{KB}$); the grating position only moves the sample. At the default 450 nm:

	pink 4 mm	mono 1 mm
grating period (solved)	5481 nm at z_g = 1.35 m	1138 nm at z_g = 1.50 m
Λ obtained	452.3 nm	450.6 nm
beams to the sample normal	∓2.58 mrad (∓0.148°)	similar
photons in A(−1)+B(+1)	0.78 % of the KB input	0.77 %
probe angle φ_B (400 nm)	26.25°	26.35°

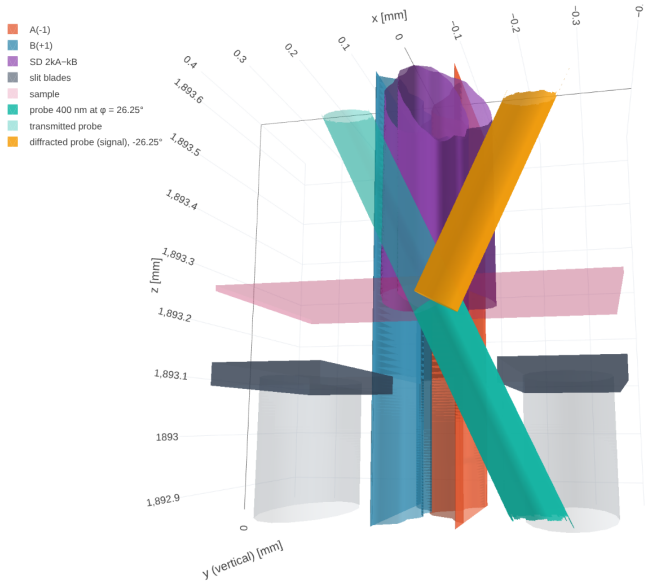
Yes, 0.78 %. An opaque amplitude grating behind a beamstop wider than the beam FWHM is not a photon-saving device. The beamstop must be that wide (1.3–1.5 × the FWHM at its position), otherwise the zero orders leak through the slit for any grating. A π phase grating gives ~4× more in the 1st orders.

The optical probe

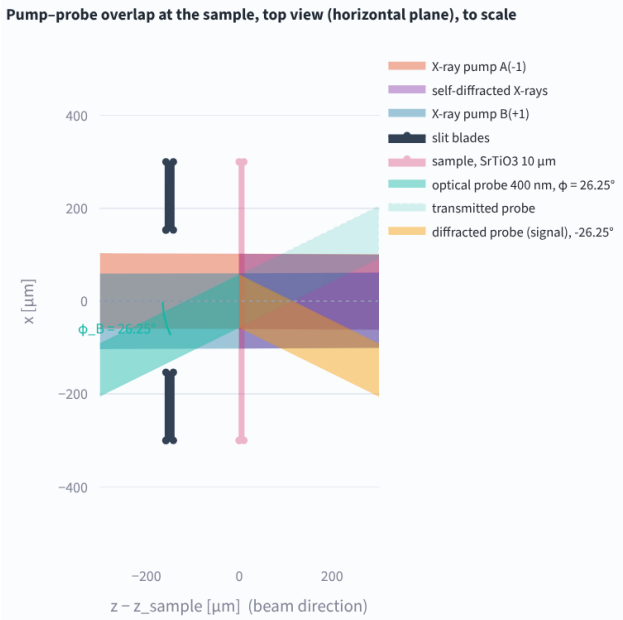
The probe is phase matched when its in-plane wavevector changes by exactly ±K: **$\sin \varphi_B = \lambda_{probe}/(2\Lambda) = (\lambda_{probe}/\lambda_X)\cdot\sin(\theta/2)$**. For 400 nm and Λ = 452 nm: $\sin \varphi_B = 0.4422$, $\varphi_B = 26.25°$ from the normal, 9.61° inside STO (n = 2.65). The diffracted probe leaves at $-\varphi_B$. With a soft-X-ray pump the grating is only $L_{att} \approx 159$ nm deep, so it is a thin (Raman–Nath) grating and the angular acceptance is huge (~129°). Being off by a few degrees is forgiven; being off by 60° (we have seen it) is not.

Three geometric constraints, all checked:

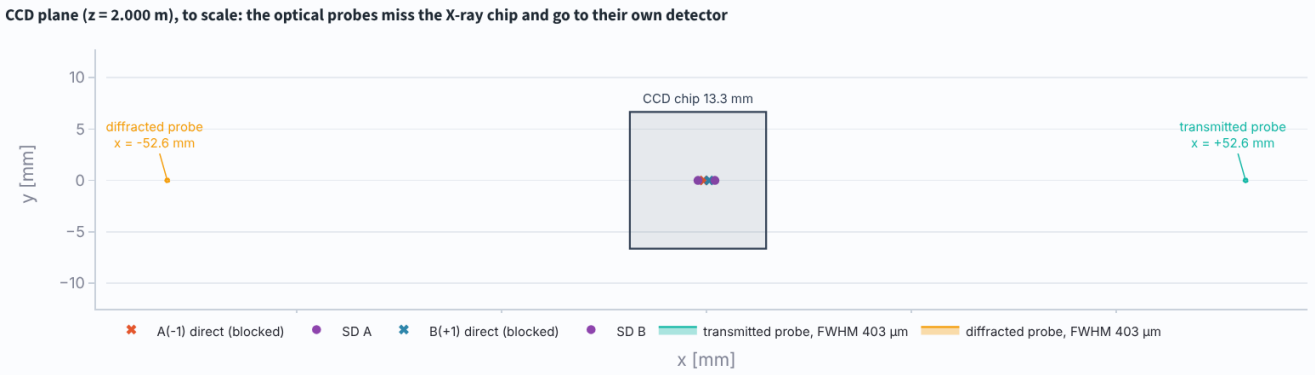
- **It must overlap the pumps.** The probe hits the sample at the X-ray spot (93 × 338 µm FWHM, pink). Its FWHM footprint (66 × 59 µm for w_0 = 50 µm) must fit inside.
- **It must pass the slit gap.** The slit blocks the unwanted X-ray orders and sits on the probe's path. Largest slit → sample distance = (gap/2 − w_0/cos φ) / tan φ: 0.20 mm pink, 0.16 mm mono. Default 0.15 mm, 100 % / 99 % of the probe passes.
- **It must stay off the X-ray CCD.** At the CCD plane (107 mm after the sample) the probes are at ±52.6 mm and ~0.4 mm wide (Rayleigh range π w_0^2/λ = 19.6 mm). The chip is 13.3 mm wide. The diffracted probe goes to a photodiode, not into our CCD.



3D close-up, to scale: A(−1), B(+1), slit blades with the gap, sample, probe at 26.25°, transmitted and diffracted probe, SD beams.



Same thing from the top in 2D, equal axes in µm, with the φ_B arc.



CCD plane to scale (mm): the chip with the X-ray spots in the middle, the transmitted and diffracted probes far outside.

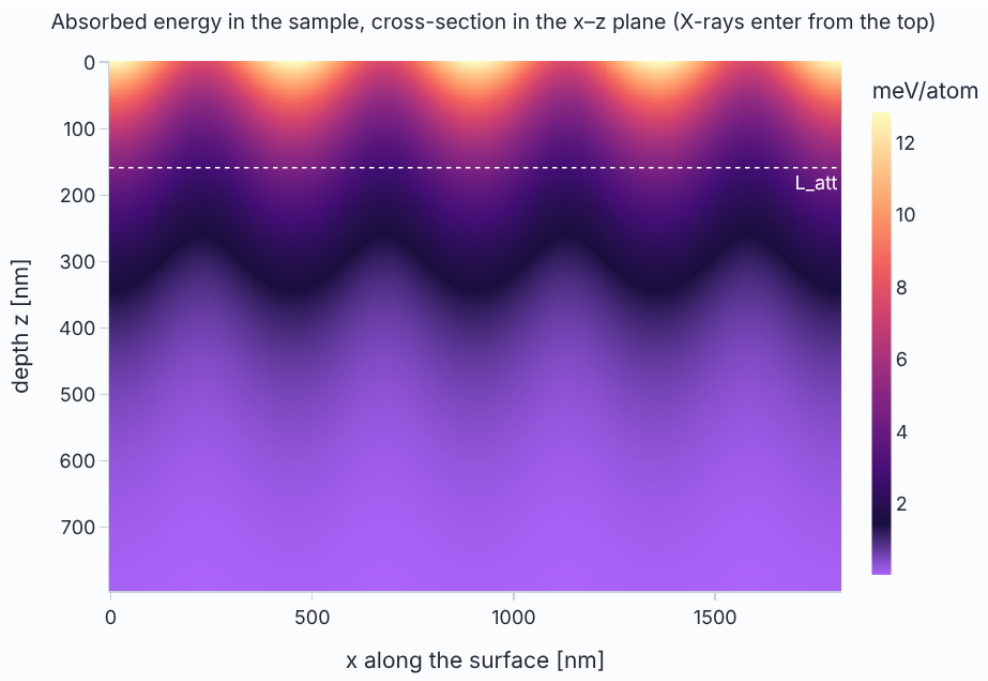
Self-diffraction and the sample

SD needs the X-rays to come out of the sample. 10 µm of $SrTiO_3$ transmits $e^{-d/L} = 4\times10^{-28}$ at 532 eV, which in round numbers is nothing. The SD signal scales as $e^{-d/L}\cdot L_{eff}^{\,2}$, with an optimum at $d = L_{att}\cdot\ln 3 \approx 174$ nm. So: SD with a membrane, optical probe with the thick crystal. The yellow warning says this every time, on purpose.

Sample material. Pick a preset ($SrTiO_3$, Si_3N_4, SiO_2, Al_2O_3, MgO, C) or type a formula and constants. Each input is compared with literature or computed values and flagged **consistent** or **not consistent**: formula (must parse), density (±5 %), Henke L_{att} (±10 %), n at the probe wavelength (±2 %), and a scan for absorption edges within ±15 eV of the pump.

For STO the edge scan finds the O K edge 12 eV above the pump. Henke tables have no near-edge structure, so the real L_{att} at 532 eV can be several times shorter than 159 nm. Put in a measured XAS value when we have one.

TG in the sample. For the pink defaults: peak absorbed energy 138 J/cm³ × (1 + V), 12.8 meV/atom at the fringe maxima, ΔT ≈ 50 K (Dulong–Petit), fringe contrast V = 0.25, everything within ~160 nm of the surface.

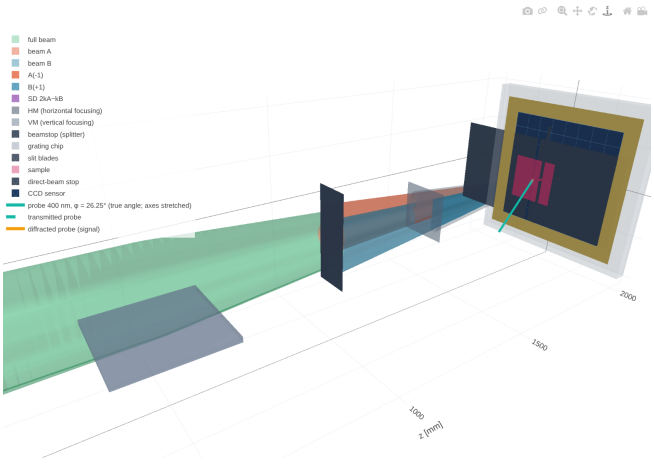


Absorbed energy in the x–z plane (X-rays from the top): 452 nm fringes, gone after a few L_{att}.

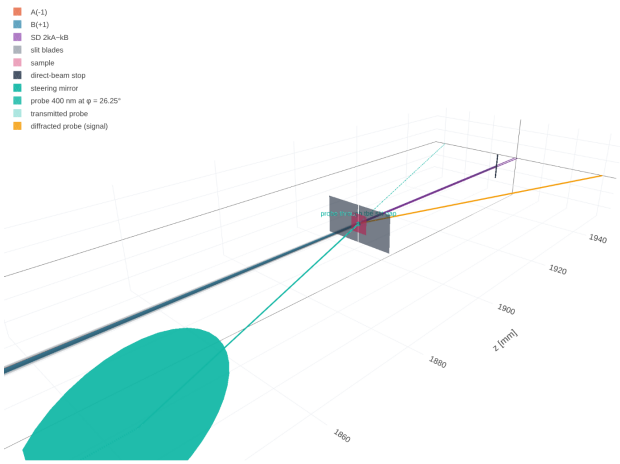
The drawings, and why we trust them now

Every view draws every beam by the same outline: its **1/e² radius**, the same definition as the w_0 we type in (2.3–97.7 % of the rays in 1D, 86.5 % in 2D). A and B are the two parts of the full beam on either side of the beamstop shadow, cut again by the slit, so they leave the full beam without a step. The FWHM would look nicer and be wrong: the beamstop is wider than the FWHM there, so A and B live entirely in the tails.

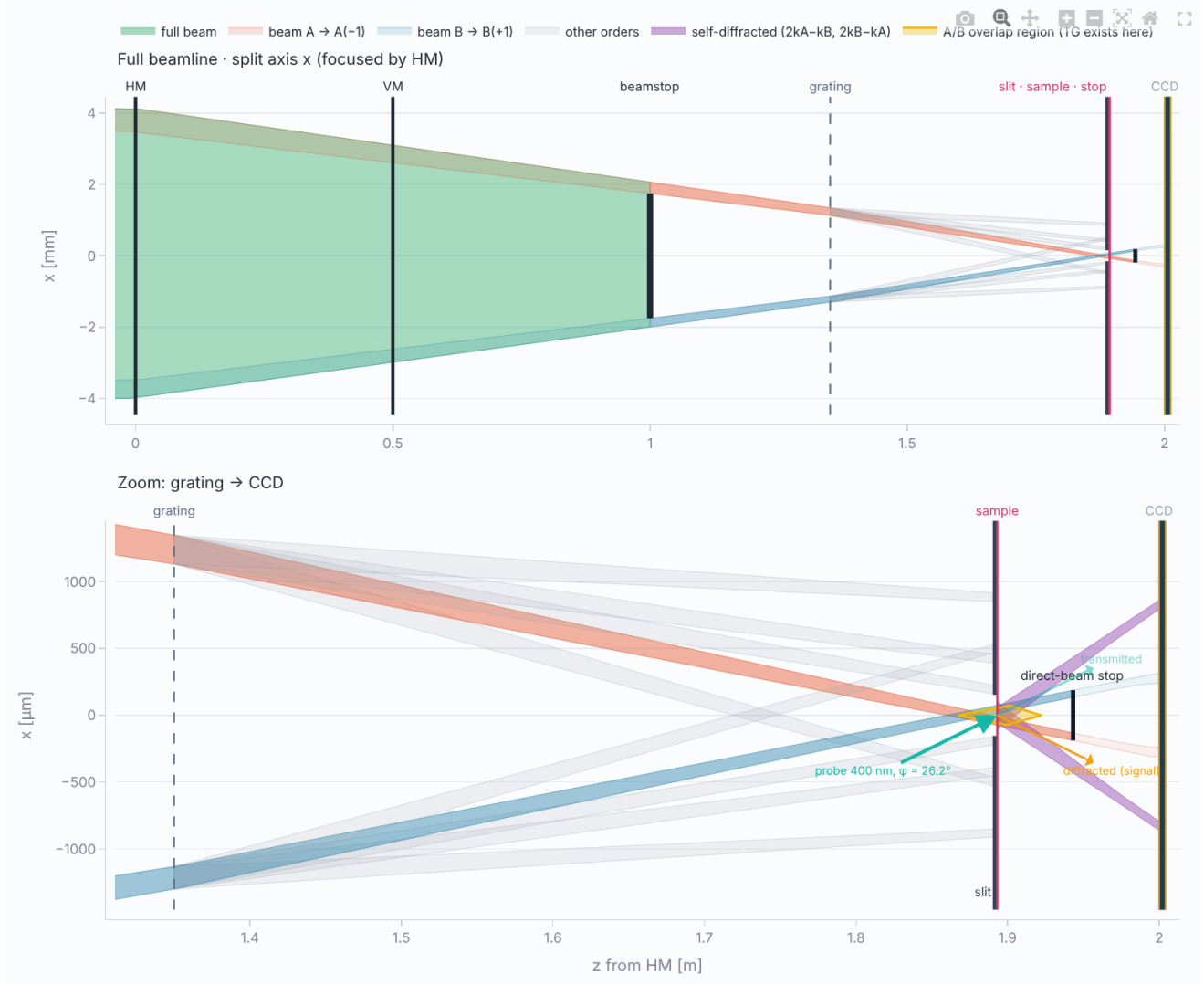
The whole-beamline views stretch the transverse axes (otherwise a 4 mm beam over 2 m is a line). There the probe is drawn through the sample at its true angle on screen, not at its stretched lab position (which would look like 89°). The to-scale views (grating → CCD, close-ups) draw everything in lab coordinates.



Whole beamline: full beam → A and B at the beamstop → orders → sample → CCD. Transverse axes stretched.



Grating → CCD, to scale: steering mirror (1″), probe through the slit gap, transmitted and diffracted probe.



Beamline tab: full sketch and grating → CCD zoom. Probe arrows at 26.25° on screen through the sample.

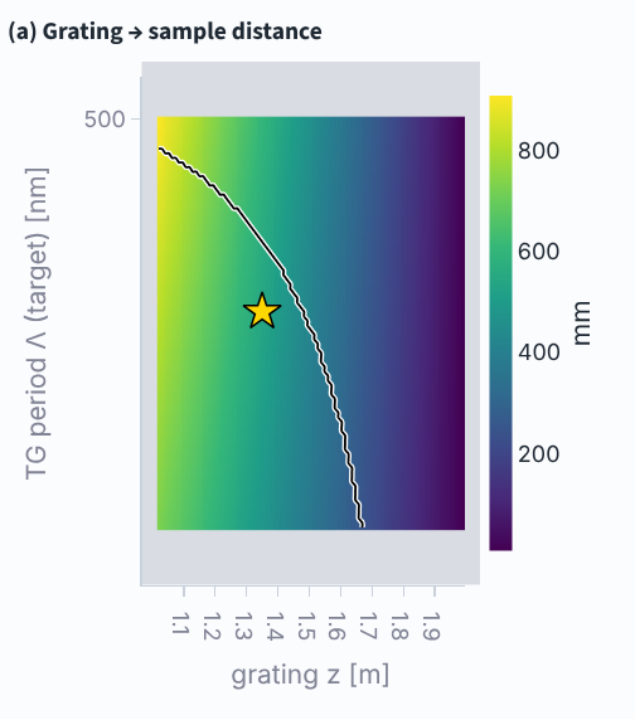
Before every release we run

```
python check_views.py
```

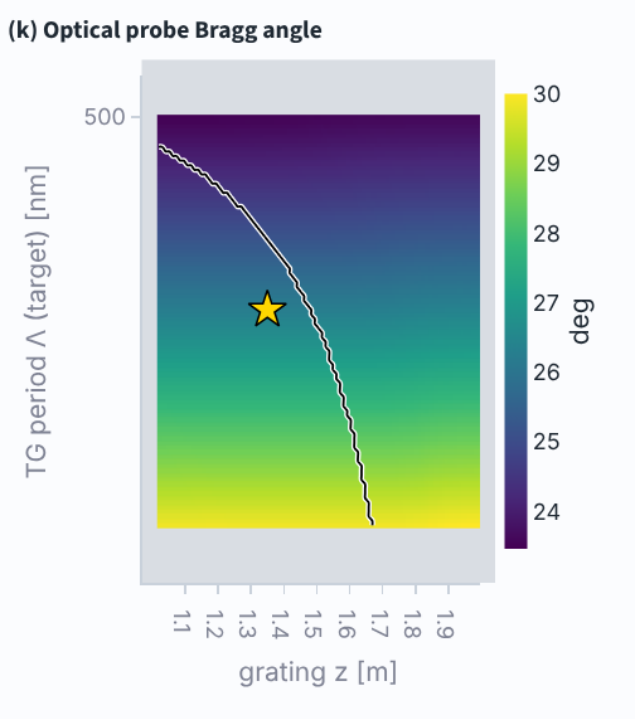
It checks, for pink and mono: the probe is at φ_B and passes the slit; its angle in all six views; A and B meet the full beam at the beamstop and their orders at the grating and the slit (2D and 3D, steps < 3 % of the beam width); the app runs and keeps the probe settings after optical → SD → optical. Exit code 0 means all passed. It already caught one of our own mistakes before it reached anybody, which is exactly the point.

Correlation scans

Each pixel is a full simulation. Everything not on the axes comes from the sidebar. The outline is the feasible region: Λ in target, sample inside the A/B overlap, ≥ 20 µm between wanted and unwanted orders at the slit, < 1 % unwanted light, overlap > 0.5, plus SD clearance and SD on the chip (SD mode) or $\Lambda > \lambda_{probe}/2$ (probe mode). The star is the current setting.



Grating → sample distance.



Phase-matched probe angle φ_B: 24–30° over the 400–500 nm target.

What we still have to put in

- Measured PAL-XFEL transverse coherence (`coh_ratio`, now 2.7), bandwidth and pulse energy (placeholders: pink 0.5 %, 100 µJ; mono 0.02 %, 10 µJ). The fluences and ΔT scale with these.
- The focal-length convention (f_V from the VM or from the HM), the real CCD chip and position.
- Measured STO absorption at 532 eV (pre-edge), and the real grating (material, bar thickness, phase).

What we want from testers

- Anything where two views disagree, a card is green when it should not be, or a number is physically odd.
- With each report: the version line from the sidebar, the saved parameter JSON (Numerics and files), a screenshot and `sdxtg_app.log`.
- Feature requests are welcome. Complaints about the 0.78 % photon efficiency go to physics, not to us.

Code: github.com/D3nFaina/xtg_and_sdxtg-simulator · MIT licence · `sdxtg_sim.py` (physics, scans, command line), `sdxtg_3d.py` (3D and close-ups), `sdxtg_wave.py` (wave optics), `sdxtg_app.py` (GUI), `run_app.py` (launcher), `check_views.py` (release checks).